

Phone: (231) 946-6767 Fax: (231) 946-8741
Email: shanna@sosanalytical.com www.sosanalytical.com

4125 Cedar Run Road, Suite B
Traverse City, MI 49684

CUSTODY TRANSFER RECORD

Client / Company Name: NA-1
Site Address: _____
Project # / WSSN #: _____
Sampling Company: NA-1
Sampler's Name: Steve Kniss
Send Results To: Colleen Allen
Address: _____
Phone: _____ Fax / Email: _____
Invoice To: NA-1
Address: _____

Quote #: _____ PO #: _____
Miscellaneous Information: _____

SOS Project ID #
160783

Cooler Temp (°C) _____ Page _____ of _____

Analysis Information

Quote #:		PO #:		Sample Information:		Sample Identification		Collection Information		Information		# of Containers		Matrix DW, MW, GV, Soil, Oil, Sludge		Comments / Other Analysis		RUSH Due: Call To Schedule																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
Miscellaneous Information:		Westford County Landfill		3/2/16		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM		AM		PM			

Relinquished by: <u>[Signature]</u>	Date: <u>3/3/16</u>	Time: <u>4:00</u>	Received by: _____	Date: _____	Time: _____
Relinquished by: _____	Date: _____	Time: _____	Received by: <u>C. Howard</u>	Date: <u>3/3/16</u>	Time: <u>4:00</u>



4125 Cedar Run Rd., Suite B
Traverse City, MI 49684
Phone 231-946-6767
Fax 231-946-8741
www.sosanalytical.com

COMPANY: NORTHERN A-1
NAME: WEXFORD COUNTY LANDFILL
PROJECT NO: 3/3/2016
WSSN: 4:00 PM
WELL PERMIT: WEXFORD COUNTY LANDFILL
TAX ID: LEACHATE
LOCATION: DATE SAMPLED: 3/2/2016
TIME SAMPLED: GRAB/WATER
COUNTRY: MI
TWP: WEXFORD

INORGANICS

Analysis	Concentration	LOD	Units	Analyst	Date Completed	Reg Limit(MCL)
NITROGEN, AMMONIA - EPA 350.1	2.70	0.30	mg/L (PPM)	KMJ	3/8/2016	

ND = NOT DETECTED
LOD = LIMIT OF DETECTION
SMCL = FEDERAL NON-ENFORCEABLE LIMIT
MCL = MAXIMUM CONTAMINANT LEVEL
s.u. = STANDARD pH UNITS REPORTED AT 25 C
DISS = DISSOLVED

APPROVED BY:

SHANNA SHEA
LAB MANAGER

[Signature]



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Traverse City, MI 49684
Phone 231-946-6767
Fax 231-946-8741
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COMPANY: NORTHERN A-1
PROJECT NO: WEXFORD COUNTY LANDFILL
NAME: WSSN:
WELL PERMIT: TAX ID:
LOCATION: MI
COUNTY:
TWP:

SOS PROJECT NO: 155848 - 1
SAMPLED BY: EDDIE ASCIONE/NA-1
DATE RECEIVED: 12/21/2015
TIME RECEIVED: 3:45 PM
SAMPLE ID: WEXFORD COUNTY LANDFILL
DATE SAMPLED: 12/21/2015
TIME SAMPLED: 11:30 AM
SAMPLE MATRIX: LEACHATE

INORGANICS

Analysis	Concentration	LOD	Units	Analyst	Completed Date	Reg Limit(MCL)
MANGANESE EPA 200.8	0.05	0.05	mg/L (PPM)	SW	12/22/2015	
NITROGEN, AMMONIA - EPA 350.1	22.1	2.0	mg/L (PPM)	KMJ	12/22/2015	
NITROGEN, NITRATE - EPA 300.0	0.70	0.30	mg/L (PPM)	KMJ	12/23/2015	
NITROGEN, NITRITE - EPA 300.0	ND	0.15	mg/L (PPM)	KMJ	12/23/2015	

ND = NOT DETECTED

LOD = LIMIT OF DETECTION

SMCL = FEDERAL NON-ENFORCEABLE LIMIT

MCL = MAXIMUM CONTAMINANT LEVEL

s.u. = STANDARD pH UNITS REPORTED AT 25 C

DISS = DISSOLVED

APPROVED BY:

SHANNA SHEA
LAB MANAGER

Shanna Shea



2105 Pless Drive · Brighton, Michigan 48114 · Phone (810) 229-7575 · Fax (810) 229-8650 · E-mail bai-brighton@sbcglobal.net

December 09, 2015

Pescador LLC
8960 Lamplighter Lane
Cadillac, MI 49601

Subject: Wexford County Landfill
040

Dear Mr. Staub :

Thank you for making Brighton Analytical, L.L.C. your laboratory of choice. Attached are the results for the samples submitted on 12/07/2015 for the above mentioned project. NELAP/TNI Accredited Analysis and MDEQ Drinking Water Certified Analysis will be identified in their respective reporting formats. Hard copies can be supplied at your request for a fee of \$20.00 per copy.

The invoice for this project will be emailed separately. If you have any questions concerning the data or invoice, please don't hesitate to contact our office. We welcome your comments and suggestions to improve our quality systems. Please reference Brighton Analytical, L.L.C. Project ID 37010 when calling or emailing. We thank you for this opportunity to partner with you on this project and hope to work with you again in the future.

Sincerely,
Brighton Analytical, L.L.C.



TABLE 4.
WEXFORD COUNTY LANDFILL
 Primary Collection System
 Monitoring Parameters and Sampling Frequency

Parameter	Active Life Sampling Frequency	Post Closure Sampling Frequency	MDL (ug/L)
PRIMARY INORGANIC INDICATOR PARAMETERS			
boron (total)	quarterly	annual	20
chloride	quarterly	annual	1,000
potassium (total)	quarterly	annual	100
sodium (total)	quarterly	annual	1,000
total inorganic nitrogen (TIN)	quarterly	annual	20
FIELD PARAMETERS			
conductivity	quarterly	annual	n/a
pH	quarterly	annual	n/a
temperature	quarterly	annual	n/a
ALTERNATE INORGANIC INDICATOR PARAMETERS			
bicarbonate alkalinity	annual	annual	10,000
calcium (total)	annual	annual	1,000*
carbonate alkalinity	annual	annual	10,000
chemical oxygen demand	annual	annual	5,000
cyanide	annual	annual	5
iron (total)	annual	annual	20
magnesium (total)	annual	annual	1,000
manganese (total)	annual	annual	5
phenolics	annual	annual	10
sulfates	annual	annual	2,000
total dissolved solids (TDS)	annual	annual	20,000
total organic carbon (TOC)	annual	annual	500
HEAVY METALS LISTED IN R299.4452			
antimony (total)	annual	annual	1
arsenic (total)	annual	annual	1
barium (total)	annual	annual	5
beryllium (total)	annual	annual	1
cadmium (total)	annual	annual	0.2
chromium (total)	annual	annual	1
cobalt (total)	annual	annual	15
copper (total)	annual	annual	1
lead (total)	annual	annual	1
nickel (total)	annual	annual	2
selenium (total)	annual	annual	1
silver (total)	annual	annual	0.2
thallium (total)	annual	annual	2
vanadium (total)	annual	annual	2
zinc (total)	annual	annual	10
PRIMARY VOCs LISTED IN R299.4453			
bromodichloromethane	quarterly	annual	1.0
bromoform; tribromomethane	quarterly	annual	1.0
carbon tetrachloride	quarterly	annual	1.0
chlorobenzene	quarterly	annual	1.0
chloroethane; ethyl chloride	quarterly	annual	5.0
chloroform; trichloromethane	quarterly	annual	1.0
dibromochloromethane; chlorodibromomethane	quarterly	annual	1.0
o-dichlorobenzene; 1,2-dichlorobenzene	quarterly	annual	1.0
p-dichlorobenzene; 1,4-dichlorobenzene	quarterly	annual	1.0
1,1-dichloroethane; ethylidene chloride	quarterly	annual	1.0
1,2-dichloroethane; ethylene dichloride	quarterly	annual	1.0
1,1-dichloroethylene; 1,1-dichloroethene	quarterly	annual	1.0
cis-1,2-dichloroethylene; cis-1,2-dichloroethene	quarterly	annual	1.0
trans-1,2-dichloroethylene; trans-1,2-dichloroethene	quarterly	annual	1.0
1,2-dichloropropane; propylene dichloride	quarterly	annual	1.0

* PER DAN STAUB, USE THESE DL'S FOR WEXFORD CO. LF - PCS RUSH PROJECT.
 DT 12/7/15

PRIMARY INORGANIC INDICATORS
 (Part 115 R 299.4450)

PARAMETER	*SUGGESTED METHOD	REPORTING LIMIT (ua/l)
CHLORIDES	325	1,000
IRON	6010B	20
SULFATES	375.1	2,000
TOTAL INORGANIC NITROGEN	353.2	20
TOTAL DISSOLVED SOLIDS	160.1	20,000

ALTERNATE INORGANIC INDICATORS
 (Part 115, R 299.4451)

PARAMETER	*SUGGESTED METHOD	REPORTING LIMIT (ua/l)
MAGNESIUM	7450	1,000
MANGANESE	6010B	5
POTASSIUM	7610	100
SODIUM	7770	1,000
BICARBONATE ALKALINITY		10,000
CARBONATE ALKALINITY		10,000
CALCIUM	7140	1,000
PHENOLICS	9066	10
CYANIDE	9010	5
TOTAL ORGANIC CARBON	415.2	500
CHEMICAL OXYGEN DEMAND	410.4	5,000
BORON	6010B	20

*EPA Methods

1) Methods for Chemical Analysis of Water and Waste 1983 2)
 SW -846

METALS
 (Part 115, R 299.4452)

PARAMETER	METHOD	REPORTING LIMIT (uQ/l)
ANTIMONY	6020	2
ARSENIC	7060A	1
BARIUM	6010B	5
BERYLLIUM	6010B	1
CADMIUM	7131A	0.2
	*6010B	*5
CHROMIUM	7191	2
	*6010B	*20
COBALT	6010B	15
COPPER	7211	2
	*6010B	*10
LEAD	7421	1
	*6010B	*50
NICKEL	6010B	25
SELENIUM	7740	2
SILVER	7761	0.5
THALLIUM	7841	2.0
VANADIUM	6010B	10
ZINC	6010B	10

• Detection limit and method for routine leachate analysis.

These limits may not be acceptable when trying to obtain a waiver from groundwater monitoring.

PRIMARY VOLATILE ORGANIC
CONSTITUENTS
(Part 115, R 299.4453)

Suggested method: SW 846 Method 8260

PARAMETER	REPORTING LIMIT (uaf)
BROMODICHLOROMETHANE	1.0
BROMOFORM TRIBROMOMETHANE	1.0
CARBON TETRACHLORIDE	1.0
CHLOROBENZENE	1.0
CHLOROETHANE ETHYL CHLORIDE	5.0
CHLOROFORM TRICHLOROMETHANE	1.0
DIBROMOCHLOROMETHANE, CHLORODIBROMOMETHANE	1.0
O-DICHLOROBENZENE 1,2-DICHLOROBENZENE	1.0
P-DICHLOROBENZENE, 1,4-DICHLOROBENZENE	1.0
1,1-DICHLOROETHANE, ETHYLIDENE CHLORIDE	1.0
1,2-DICHLOROETHANE, ETHYLENE DICHLORIDE	1.0
1,1-DICHLOROETHYLENE 1,1-DICHLOROETHENE	1.0
CIS-1,2-DICHLOROETHYLENE, CIS-1,2-DICHLOROETHENE	1.0
TRANS-1,2-DICHLOROETHYLENE, TRANS-1,2-DICHLOROETHENE	1.0
1,2-DICHLOROPROPANE PROPYLENE DICHLORIDE	1.0
CIS-1,3-DICHLOROPROPENE	1.0
TRANS-1,3-DICHLOROPROPENE	1.0
METHYL BROMIDE BROMOMETHANE	5.0
METHYL CHLORIDE CHLOROMETHANE	5.0
METHYLENE BROMIDE, DIBROMOMETHANE	1.0
METHYLENE CHLORIDE, DICHLOROMETHANE	5.0
METHYL IODIDE, IODOMETHANE	1.0
1,1,1,2-TETRACHLOROETHANE	1.0
1,1,1,2,2-TETRACHLOROETHANE	1.0
TETRACHLOROETHYLENE, TETRACHLOROETHENE, PERCHLOROETHYLENE	1.0
1,1,1-TRICHLOROETHANE METHYL CHLOROFORM	1.0
1,1,2-TRICHLOROETHANE	1.0
TRICHLOROETHYLENE, TRICHLOROETHENE	1.0
TRICHLOROFLUOROMETHANE	1.0
1,2,3-TRICHLOROPROPANE	1.0
VINYL CHLORIDE	1.0
BENZENE	1.0
ETHYL BENZENE	1.0
STYRENE	1.0
TOLUENE	1.0
XYLENES	2.0

SECONDARY ORGANIC CONSTITUENTS
(Part 115, R 299.4454)

Suggested method: SW 846 Method 8260

PARAMETER	REPORTING LIMIT (uQII)
ACETONE	20.0
ACRYLONITRILE	5.0
BROMOCHLOROMETHANE	1.0
CARBON DISULFIDE	1.0
1,2-DIBROMO-3-CHLOROPROPANE, DBCP	5.0
1,2-DIBROMOETHANE ETHYLENE DIBROMIDE EDB	1.0
METHYL ETHYL KETONE, 2-BUTANONE	5.0
4-METHYL-2-PENTANONE, METHYL ISOBUTYL KETONE	5.0
TRANS-1,4-DICHLORO-2-BUTENE	1.0
2-HEXANONE, METHYL BUTYL KETONE	5.0

1,1,1,2-Tetrachloroethane	1	\$VOLLAND
1,1,1-Trichloroethane	1	\$VOLLAND
1,1,2,2-Tetrachloroethane	1	\$VOLLAND
1,1,2-Trichloroethane	1	\$VOLLAND
1,1-Dichloroethane	1	\$VOLLAND
1,1-Dichloroethene	1	\$VOLLAND
1,2,3-Trichloropropane	1	\$VOLLAND
1,2-Dibromo-3-chloropropane	5	\$VOLLAND
1,2-Dibromoethane	1	\$VOLLAND
1,2-Dichlorobenzene	1	\$VOLLAND
1,2-Dichloroethane	1	\$VOLLAND
1,2-Dichloropropane	1	\$VOLLAND
1,4-Dichlorobenzene	1	\$VOLLAND
2-Hexanone	5	\$VOLLAND
4-Methyl-2-Pentanone(MIBK)	5	\$VOLLAND
Acetone	20	\$VOLLAND
Acrylonitrile	5	\$VOLLAND
Benzene	1	\$VOLLAND
Bromochloromethane	1	\$VOLLAND
Bromodichloromethane	1	\$VOLLAND
Bromoform	1	\$VOLLAND
Bromomethane/Methyl bromide	5	\$VOLLAND
Carbon disulfide	1	\$VOLLAND
Carbon tetrachloride	1	\$VOLLAND
Chlorobenzene	1	\$VOLLAND
Chloroethane/Ethyl chloride	5	\$VOLLAND
Chloroform	1	\$VOLLAND
Chloromethane/Methyl chloride	5	\$VOLLAND
cis-1,2-Dichloroethene	1	\$VOLLAND
cis-1,3-Dichloropropene	1	\$VOLLAND
Dibromochloromethane	1	\$VOLLAND
Dibromomethane(Methylene bromide)	1	\$VOLLAND
Ethyl benzene	1	\$VOLLAND
Methyl Ethyl Ketone/2 Butanone	5	\$VOLLAND
Methyl Iodide	1	\$VOLLAND
Methylene Chloride	5	\$VOLLAND
Styrene	1	\$VOLLAND
Tetrachloroethene	1	\$VOLLAND
Toluene	1	\$VOLLAND
trans-1,2-Dichloroethene	1	\$VOLLAND
trans-1,3-Dichloropropene	1	\$VOLLAND
Trans-1,4,-Dichloro-2-Butene	5	\$VOLLAND
Trichloroethene	1	\$VOLLAND
Trichlorofluoromethane	5	\$VOLLAND
Vinyl chloride	5	\$VOLLAND

Xylenes	2	\$VOLLAND
Chloride	1000	CHLORIDE
Total Iron	20	FE602TTB
Sulfates	2000	SULFLAND
Total Inorganic Nitrogen(calc)	20	TIN
Ammonia	20	NH4-RICH
Nitrate	20	NO3-RICH
Nitrite	20	NO2-RICH
Total Dissolved Solids	20000	TDS-LAND
Total Magnesium	1000	MG602TTB
Total Manganese	5	MN60T5
Total Potassium	100	K6020WT
Total Sodium	1000	NA6020WT
Bicarbonate Alkalinity	10000	ALKBIC10
Carbonate Alkalinity	10000	ALK-CARB
Total Calcium	1000	CA602TTB
Phenols	10	PHEN4TAY
Total Cyanide	5	CYANLAND
Total Organic Carbon	500	TOC-LAND
Chemical Oxygen Demand	5000	COD-LAND
Total Boron	20	BO601TTB
Total Antimony	2	SB60T2PPB
Total Arsenic	1	AS6020WT
Total Barium	5	BA602TTB
Total Beryllium	1	BE602TTB
Total Cadmium	0.2	CD6020WT
Total Chromium	2	CR602TTB
Total Cobalt	15	CO6020WT
Total Copper	2	CU602TTB
Total Lead	1	PB602TTB
Total Nickel	25	NI6020T25
Total Selenium	2	SE60T2
Total Silver	0.5	AG6020WT
Total Sodium	1000	NA6020WT
Total Thallium	2	TL6020WT
Total Vanadium	10	VA6020WT
Total Zinc	10	ZN602T10

Subject: RE: Wexford Co LF COC
From: Daniel Staub (dmstaub@pescadorllc.com)
To: bai-brighton@sbcglobal.net;
Date: Monday, December 7, 2015 8:35 PM

Here you go, please run the blank and report with the Composite sample.

Thanks

Dan STaub

From: Brighton Analytical [mailto:bai-brighton@sbcglobal.net]
Sent: Monday, December 07, 2015 4:37 PM
To: Daniel Staub <dmstaub@pescadorllc.com>
Subject: Wexford Co LF COC

Hi Dan,

Here is the COC for the samples that Rick drove down. Michelle has listed the "Blank" VOAs that were not on the COC as 1 Trip Blank with 2 VOA vials. If you would like them listed separately and/or tested separately, please mark up the COC and we will accommodate.

Thanks,

Colleen

Brighton Analytical, LLC

2105 Pless Dr.
Brighton, MI 48114
(810) 229-7575
www.brightonanalytical.net



Brighton Analytical, L.L.C.
2105 Pless Drive
Brighton, Michigan 48116
Phone: (810) 229-7575 FAX: (810) 229-8650
e-mail: bai-brighton@sbcglobal.net

To: Pescador LLC
8960 Lamplighter Lane
Cadillac, MI 49601

Sample Date: 12/7/2015
Submit Date: 12/7/2015
Report Date: 12/9/2015

BA Report Number: 37010
BA Sample ID: CC03376

Project Name: Wexford County Landfill
Project Number: 040
Sample ID: Composite 1

Parameters	Results	Units	DL	Method Reference	Analyst	Analysis Date
Total Metal Analysis						
Total Antimony	Not detected	ug/L	2	SW846 6020	LT	12/8/2015
Total Arsenic	1	ug/L	1	SW846 6020	LT	12/8/2015
Total Barium	64	ug/L	5	SW846 6020	LT	12/8/2015
Total Beryllium	Not detected	ug/L	1	SW846 6020	LT	12/8/2015
Total Boron	130	ug/L	20	SW846 6010B	LT	12/9/2015
Total Cadmium	Not detected	ug/L	0.2	SW846 6020	LT	12/8/2015
Total Calcium	26000	ug/L	1000	SW846 6020	LT	12/8/2015
Total Chromium	3	ug/L	2	SW846 6020	LT	12/8/2015
Total Cobalt	Not detected	ug/L	15	SW846 6020	LT	12/8/2015
Total Copper	4	ug/L	2	SW846 6020	LT	12/8/2015
Total Iron	190	ug/L	20	SW846 6020	LT	12/8/2015
Total Lead	Not detected	ug/L	1	SW846 6020	LT	12/8/2015
Total Magnesium	17000	ug/L	1000	SW846 6020	LT	12/8/2015
Total Manganese	78	ug/L	5	SW846 6020	LT	12/8/2015
Total Nickel	Not detected	ug/L	25	SW846 6020	LT	12/8/2015
Total Potassium	86000	ug/L	100	SW846 6020	LT	12/8/2015
Total Selenium	Not detected	ug/L	2	SW846 6020	LT	12/8/2015
Total Silver	Not detected	ug/L	0.5	SW846 6020	LT	12/8/2015
Total Sodium	110000	ug/L	1000	SW846 6020	LT	12/8/2015
Total Thallium	Not detected	ug/L	2	SW846 6020	LT	12/8/2015
Total Vanadium	Not detected	ug/L	10	SW846 6020	LT	12/8/2015
Total Zinc	20	ug/L	10	SW846 6020	LT	12/8/2015
Metal Water Total (digest)	Digested			3015	LT	12/8/2015
Inorganic Analysis						
Ammonia	21000	ug/L	20	SM4500 NH3G	RM	12/7/2015
Bicarbonate Alkalinity	300000	ug/L	10000	SM2320B	LS	12/8/2015
Carbonate Alkalinity	Not detected	ug/L	10000	SM2320B	LS	12/8/2015
Chloride	190000	ug/L	1000	EPA 300.0R2.1	RM	12/7/2015
Nitrate	380	ug/L	20	EPA 300.0/4500	RM	12/7/2015



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To: Pescador LLC
8960 Lamplighter Lane
Cadillac, MI 49601

Sample Date: 12/7/2015
Submit Date: 12/7/2015
Report Date: 12/9/2015

BA Report Number: 37010
BA Sample ID: CC03376

Project Name: Wexford County Landfill
Project Number: 040
Sample ID: Composite 1

Parameters	Results	Units	DL	Method Reference	Analyst	Analysis Date
Nitrite	90	ug/L	20	EPA 300.0/4500	RM	12/7/2015
Sulfates	22000	ug/L	2000	EPA 300.0R2.1	RM	12/7/2015
Total Cyanide	5	ug/L	5	SW846 9012A	RM	12/8/2015
Total Dissolved Solids	600000	ug/L	20000	SM2540C	LS	12/8/2015
Total Inorganic Nitrogen(calc)	21000	ug/L	20		RM	12/7/2015

Organic Analysis

Chemical Oxygen Demand	40000	ug/L	5000	EPA 410.4	MB	12/8/2015
Phenols	Not detected	ug/L	10	SM5530D	RM	12/8/2015
Total Organic Carbon	19000	ug/L	500	SM5310B	RG	12/8/2015

Volatile Analysis

Acetone	Not detected	ug/L	20	SW846 8260B	CW	12/9/2015
Acrylonitrile	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Benzene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Bromochloromethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Bromodichloromethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Bromoform	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Bromomethane/Methyl bromide	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Carbon disulfide	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Carbon tetrachloride	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Chlorobenzene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Chloroethane/Ethyl chloride	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Chloroform	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Chloromethane/Methyl chloride	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
cis-1,2-Dichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
cis-1,3-Dichloropropene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,2-Dibromo-3-chloropropane	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Dibromochloromethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,2-Dibromoethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Dibromomethane/Methylene bromide	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,2-Dichlorobenzene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015



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To: Pescador LLC
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Cadillac, MI 49601

Sample Date: 12/7/2015
Submit Date: 12/7/2015
Report Date: 12/9/2015

BA Report Number: 37010
BA Sample ID: CC03376

Project Name: Wexford County Landfill
Project Number: 040
Sample ID: Composite 1

Parameters	Results	Units	DL	Method Reference	Analyst	Analysis Date
1,4-Dichlorobenzene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,1-Dichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,2-Dichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,1-Dichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,2-Dichloropropane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Ethyl benzene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
2-Hexanone	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Methyl Ethyl Ketone/2 Butanone	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Methyl Iodide	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Methyl(tert)butyl ether (MTBE)	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
4-Methyl-2-Pentanone(MIBK)	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Methylene Chloride(Dichloromethane)	Not detected	ug/L	5	SW846 8260B	CW	12/9/2015
Styrene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,1,1,2-Tetrachloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,1,2,2-Tetrachloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Tetrachloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Toluene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
trans-1,2-Dichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
trans-1,3-Dichloropropene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
trans-1,4,-Dichloro-2-Butene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,1,1-Trichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,1,2-Trichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Trichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Trichlorofluoromethane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
1,2,3-Trichloropropane	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Vinyl chloride	Not detected	ug/L	1	SW846 8260B	CW	12/9/2015
Xylenes	Not detected	ug/L	2	SW846 8260B	CW	12/9/2015



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Submit Date: 12/7/2015
Report Date: 12/9/2015

To: Pescador LLC
8960 Lamplighter Lane
Cadillac, MI 49601

BA Report Number: 37010
BA Sample ID: CC03376

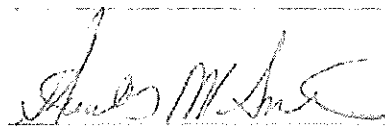
Project Name: Wexford County Landfill
Project Number: 040
Sample ID: Composite 1

Parameters	Results	Units	DL	Method Reference	Analyst	Analysis Date
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DL=Reported detection limit for analytical method requested. Some compounds require special analytical methods to achieve MDNR designated target detection limits (TDL).

Released by:

Date:


09 DEC 2015



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Cadillac, MI 49601

Sample Date: 12/7/2015
Submit Date: 12/7/2015
Report Date: 12/9/2015

BA Report Number: 37010
BA Sample ID: CC03378

Project Name: Wexford County Landfill
Project Number: 040
Sample ID: Trip Blank

Parameters	Results	Units	DL	Method Reference	Analyst	Analysis Date
Volatile Analysis						
Acetone	Not detected	ug/L	20	SW846 8260B	CW	12/8/2015
Acrylonitrile	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Benzene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Bromochloromethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Bromodichloromethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Bromoform	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Bromomethane/Methyl bromide	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Carbon disulfide	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Carbon tetrachloride	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Chlorobenzene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Chloroethane/Ethyl chloride	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Chloroform	14	ug/L	1	SW846 8260B	CW	12/8/2015
Chloromethane/Methyl chloride	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
cis-1,2-Dichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
cis-1,3-Dichloropropene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,2-Dibromo-3-chloropropane	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Dibromochloromethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,2-Dibromoethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Dibromomethane/Methylene bromide	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,2-Dichlorobenzene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,4-Dichlorobenzene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,1-Dichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,2-Dichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,1-Dichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,2-Dichloropropane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Ethyl benzene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
2-Hexanone	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Methyl Ethyl Ketone/2 Butanone	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Methyl Iodide	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Methyl(tert)butyl ether (MTBE)	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015



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Cadillac, MI 49601

Sample Date: 12/7/2015
Submit Date: 12/7/2015
Report Date: 12/9/2015

BA Report Number: 37010
BA Sample ID: CC03378

Project Name: Wexford County Landfill
Project Number: 040
Sample ID: Trip Blank

Parameters	Results	Units	DL	Method Reference	Analyst	Analysis Date
4-Methyl-2-Pentanone(MIBK)	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Methylene Chloride(Dichloromethane)	Not detected	ug/L	5	SW846 8260B	CW	12/8/2015
Styrene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,1,1,2-Tetrachloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,1,2,2-Tetrachloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Tetrachloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Toluene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
trans-1,2-Dichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
trans-1,3-Dichloropropene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
trans-1,4,-Dichloro-2-Butene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,1,1-Trichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,1,2-Trichloroethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Trichloroethene	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Trichlorofluoromethane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
1,2,3-Trichloropropane	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Vinyl chloride	Not detected	ug/L	1	SW846 8260B	CW	12/8/2015
Xylenes	Not detected	ug/L	2	SW846 8260B	CW	12/8/2015

DL=Reported detection limit for analytical method requested. Some compounds require special analytical methods to achieve MDNR designated target detection limits (TDL).

Released by:

Date: 04 DEC 2015



BRIGHTON ANALYTICAL, LLC

QUALITY ASSURANCE/QUALITY
CONTROL

REPRESENTATIVE BATCH QUALITY CONTROL

Accuracy & Precision

Analyst: LS

Parameter: Alkalinity

Analysis Date: 12/08/15

Method Reference: 310.1 / SM2320B

Matrix: Total

Batch: W1

SPIKE - ACCURACY					
Laboratory ID	Spike Concentration (ug/L)	Background (ug/L)	Recoveries (%)	Acceptable Range (%)	Method Blank Concentration (ug/L)
CC003376	250000	301000	95 / 95	80 - 120	<5000
SPIKE - PRECISION					
Laboratory ID	Observed A (ug/L)	Observed B (ug/L)	RPD (%)	Acceptable Range	
CC003376	537500	537500	0.0	0 - 20	
MISCELLANEOUS					
	Standard ID #	(%)	Acceptable Range		
Independent Secondary Reference Material:	P239-506	99	90 - 110		
Method Standard (Laboratory Control Spike):	09302015LS	116	80 - 120		

COMMENTS: _____

REPRESENTATIVE BATCH QUALITY CONTROL

Accuracy & Precision

Analyst: RM

Parameter: AMMONIA

Analysis Date: 12/7/15

Method Reference: SM4500 NH3G

SPIKE - ACCURACY

Laboratory Identification	Spike Conc. (µg/L)	Background (µg/L)	Percent Recoveries	Acceptable Range (%)	Method Blank Concentration
3255 MS/MSD	500	60	110 / 104	80 - 120	< 10 ppb

SPIKE - PRECISION

Laboratory Identification	Observed A (µg/L)	Observed B (µg/L)	RPD	Acceptable Range	
3255 MS/MSD	608	582	4.4%	≤ 20%	

MISCELLANEOUS

	Standard ID #	%Recovery	
Independent Secondary Reference Material:	WP 240	103%	

COMMENTS: _____

REPRESENTATIVE BATCH QUALITY CONTROL

Accuracy & Precision

Analyst: MB

Parameter: COD

Analysis Date: 12/8/2015

Method Reference: EPA 410.4

SPIKE - ACCURACY

Laboratory ID	Spike Concentration	Background	% Recoveries	Range (%)	Method Blank Concentration
CC3376	100000PPB	20000	105/100	90-110	<3000

SPIKE - PRECISION

Laboratory ID	Observed A	Observed B	RPD	Acceptable Range	
CC3376	125000	120000	4.1	80-120	

MISCELLANEOUS

	Standard ID #	% Recovery	
Independent Secondary Reference Material:	PHENOVA 8135-07	104	
Method Standard (Laboratory Control Spike):			

COMMENTS: _____

Representative Batch Precision And Accuracy Quality Control Summary

Ion Chromatograph EPA Method 300.0

Date: 12/7/15Reviewed by: SSAnalyst: RM

ERA P239

ERA # : WP 240Exp. Date: Apr-18

Analyte	Sample Conc	LCS Value	LCS Conc.	% Rec. LCS	ERA Conc.	ERA TRUE Value	% Rec ERA	Control limits	Units
Fluoride	<100	5,000	4,879	98	3,106	3,290	94	90-110%	ug/L
Chloride	<1000	50,000	49,204	98	51,029	52,300	98	90-110%	ug/L
Nitrite	<10	1,000	944	94				90-110%	ug/L
Nitrate	<10	1,000	1,053	105	14,058	14,000	100	90-110%	ug/L
Sulfate	<1000	50,000	46,088	92	34,354	37,600	91	90-110%	ug/L

Sample ID# 3365

Analyte	Sample Conc	Spike Value	MS Conc.	MSD Conc.	% Rec MS	% Rec MSD	RPD	Control limits	Units
Fluoride	416	5,000	5,445	5,027	101	92	8.0	80-120%	ug/L
Chloride	172,795	50,000	226,373	223,988	107	102	1.1	80-120%	ug/L
Nitrite	0	1,000	977	936	98	94	4.3	80-120%	ug/L
Nitrate	120	1,000	871	823	75	70	5.7	80-120%	ug/L
Sulfate	37,687	50,000	86,035	84,288	97	93	2.1	80-120%	ug/L

Nitrate recovery is out of control due to sample matrix.

Sample ID# 3064

Analyte	Sample Conc	Spike Value	MS Conc.	MSD Conc.	% Rec MS	% Rec MSD	RPD	Control limits	Units
Fluoride	0	5,000	4,673	4,517	93	90	3.4	80-120%	ug/L
Chloride	2,651	50,000	53,886	52,282	102	99	3.0	80-120%	ug/L
Nitrite	0	1,000	1,002	976	100	98	2.7	80-120%	ug/L
Nitrate	0	1,000	941	894	94	89	5.1	80-120%	ug/L
Sulfate	1,690	50,000	51,825	50,573	100	98	2.4	80-120%	ug/L

Sample ID# 3074

Analyte	Sample Conc	Spike Value	MS Conc.	MSD Conc.	% Rec MS	% Rec MSD	RPD	Control limits	Units
Fluoride	0	5,000	4,640	4,536	93	91	2.3	80-120%	ug/L
Chloride	2,272	50,000	52,988	51,311	101	98	3.2	80-120%	ug/L
Nitrite	57	1,000	1,017	988	96	93	2.9	80-120%	ug/L
Nitrate	44	1,000	935	902	89	86	3.6	80-120%	ug/L
Sulfate	549	50,000	55,821	53,074	111	105	5.0	80-120%	ug/L

REPRESENTATIVE BATCH QUALITY CONTROL

Accuracy & Precision

Analyst: RM

Parameter: Cyanide

Analysis Date: 12/8/15

Method Reference: 335.1/335.4/9010

SPIKE - ACCURACY					
Laboratory Identification	Spike Conc. (µg/L)	Background (µg/L)	Percent Recoveries	Acceptable Range (%)	Method Blank Concentration
3376 MS/MSD	200	ND	98 / 97	90 - 110	<5
SPIKE - PRECISION					
Laboratory Identification	Observed A (µg/L)	Observed B (µg/L)	RPD	Acceptable Range	
3376 MS/MSD	195	194	0.5	≤ 20%	
MISCELLANEOUS					
	Standard ID #	%Recovery			
Independent Secondary Reference Material:	ERA P229	97%			
Method Standard (Laboratory Control Spike):		98%			

COMMENTS: _____

REPRESENTATIVE BATCH QUALITY CONTROL

Accuracy & Precision

Analyst: RM

Parameter: Phenols

Analysis Date: 12/8/15

Method Reference: 420.1

SPIKE - ACCURACY					
Laboratory ID	Spike Concentration	Background	% Recoveries	Acceptable Range (%)	Method Blank Concentration
3282 MS/MSD	500 ppb	ND	93/97	80 - 120	< 10

SPIKE - PRECISION					
Laboratory ID	Observed A	Observed B	RPD	Acceptable Range	
3282 MS/MSD	553	577	4.24%	≤20%	

MISCELLANEOUS			
	Standard ID #	% Recovery	
Independent Secondary Reference Material:	ERA P229	95%	
Method Standard (Laboratory Control Spike):		101%	

COMMENTS: _____

REPRESENTATIVE BATCH QUALITY CONTROL

Accuracy & Precision

Analyst: LS

Parameter: TDS

Analysis Date: 12/8/2015

Method Reference: SM2540C

SPIKE - ACCURACY

Laboratory ID	Spike Concentration	Recovered	% Recoveries	Acceptable Range (%)	Method Blank Concentration
PHE 8153-09	543000	512000	95	80-120	<10000

SPIKE - PRECISION

Laboratory ID	Observed A	Observed B	RPD	Acceptable Range	
CC03319	1367000	1370000	0.1	≤ 20%	

	Standard ID #	% Recovery	
Independent Secondary Reference Material:			
Method Standard (Laboratory Control Spike):			

COMMENTS: _____

REPRESENTATIVE BATCH QUALITY CONTROL

Accuracy & Precision

Analyst: RG

Parameter: TOC

Analysis Date: 12/8/2015

Method Reference: EPA 415.1/SM5310B/9060

SPIKE - ACCURACY					
Laboratory ID	Spike level PPB	Background PPB	Recoveries (%)	Acceptable Range (%)	Method Blank Concentration
CC03255	TV=10000	2300	88/91	80 - 120	ND
SPIKE - PRECISION					
Laboratory ID	Observed A PPB	Observed B PPB	RPD (%)	Acceptable Range(%)	
CC03255	11100	11400	2.70	≤ 20	
MISCELLANEOUS					
		Standard ID #	%Recoveries		
Independent Secondary Reference Material:		WP-8135-07	104		
Method Standard (Lab. Control Spike):		#1647.11	99		

COMMENTS: _____

GC/MS

VOLATILE METHOD 8260/624

REPRESENTATIVE BATCH PRECISION AND ACCURACY QUALITY CONTROL SUMMARY

Analysis Date:	December 8, 2015	Spike Std. ID:	2537.1	Inst./Detec:	HP7890/5975 Vol5
Laboratory ID:	CC03216-17	Matrix:	Water	Analyst:	CW

	Matrix Spike - Precision			Matrix spike - Accuracy				Sample background ug/L	LCS +/- 20%	Method Blank
	Spike 1 ug/L	Spike 1 ug/L	Relative Percent Difference +/- 20%	Spk Conc ug/L	% Recovery	% Recovery	Range (%)			
d4-1,2-Dichloroethane	50.1	49.4	1.2	50	100	99	70-130	100%	99%	96%
d8-Toluene	50.8	48.8	4.0	50	102	98	70-130	97%	98%	98%
4-Bromofluorobenzene	48.7	49.2	1.1	50	97	98	70-130	109%	99%	108%
COMPOUNDS										
1,1- Dichloroethane	56.9	52.3	8.5	50	114%	105%	70-130	<1	103%	<1
1,1- Dichloroethane	58.0	52.7	9.6	50	116%	105%	70-130	<1	111%	<1
Chloroform	56.2	51.2	9.3	50	112%	102%	70-130	<1	104%	<1
1,2- Dichloroethane	53.4	48.8	8.8	50	107%	98%	70-130	<1	106%	<1
Carbon tetrachloride	57.7	51.5	11.3	50	115%	103%	70-130	<1	99%	<1
1,2- Dichloropropane	55.4	50.2	9.9	50	111%	100%	70-130	<1	108%	<1
cis- 1,3 -dichloropropene	53.1	46.9	12.4	50	108%	94%	70-130	<1	105%	<1
Trichloroethene	62.2	58.6	9.4	50	124%	113%	70-130	<1	108%	<1
Dibromochloromethane	53.1	47.7	10.7	50	106%	95%	70-130	<1	109%	<1
1,1,2- Trichloroethane	54.5	48.9	10.9	50	109%	98%	70-130	<1	103%	<1
Tetrachloroethane	52.6	47.8	9.6	50	105%	96%	70-130	<1	107%	<1
1,1,2,2 -Tetrachloroethane	54.0	48.5	10.8	50	108%	97%	70-130	<1	101%	<1
Chlorobenzene	55.4	52.1	6.1	50	111%	104%	70-130	<1	106%	<1
1,3- Dichlorobenzene	54.1	51.0	5.8	50	108%	102%	70-130	<1	106%	<1
1,4- Dichlorobenzene	52.2	48.0	8.3	50	104%	96%	70-130	<1	105%	<1
1,2-Dichlorobenzene	53.5	49.3	8.1	50	107%	99%	70-130	<1	105%	<1

ug/L is equivalent to ppb

Comments: _____

ICP-OES METHOD 6010

REPRESENTATIVE BATCH PRECISION AND ACCURACY QUALITY CONTROL SUMMARY

Analysis Date: 12/9/2015

Standard ID: 112315 ICP

Batch: 12/7/2015

Matrix Spike Lab ID: CC03376

Matrix: Total

Analyst: LT

Metals	Matrix Spike - Precision *			Matrix Spike - Accuracy**				Miscellaneous***		
	Matrix Spike (ug/L)	Matrix Spike Dup (ug/L)	RPD (%)	Spk Conc (ug/L)	MS Recovery (%)	MSD Recovery (%)	Sample Conc (ug/L)	Method Blk (ug/L)	LCS-Method STD (%)	Ind. Std. SPEX 1&3 (%)
Boron	477	466	2.3	400	86.8	84.0	130	<20	86.7	94.8

* Matrix spike precision range +/- 20% RPD

** Matrix spike accuracy range +/- 30% recovery

*** LCS accuracy range +/- 15% recovery / Ind std accuracy range +/- 10% recovery

Comments: _____

ICP-MS

EPA METHOD 200.8/6020

REPRESENTATIVE BATCH PRECISION AND ACCURACY QUALITY CONTROL SUMMARY

Analysis Date: 12/8/2015

Standard ID: 112315 H2O

Batch: 12/8/2015

Matrix Spike Lab ID: CC03364

Matrix: Total

Analyst: LT

Metals	Matrix Spike - Precision *			Matrix Spike - Accuracy**				Miscellaneous***		
	Matrix Spike (ug/L)	Matrix Spike Dup (ug/L)	RPD (%)	Spk Conc (ug/L)	MS Recovery (%)	MSD Recovery (%)	Sample Conc (ug/L)	Method Blk (ug/L)	LCS-Method STD (%)	Ind. Std. SPEX 1&3 (%)
Beryllium	2065	2037	1.4	2000	103.3	101.9	0	<1	100.3	96.5
Sodium	27102	27097	0.0	20000	100.7	100.7	6954	<1000	102.3	96.5
Magnesium	38644	38634	0.0	20000	94.8	94.8	19677	<100	97.9	93.2
Aluminium	2077	2038	1.9	2000	103.9	101.9	0	<50	101.5	106.9
Potassium	21467	21187	1.3	20000	100.8	99.4	1300	<100	99.9	96.5
Calcium	91962	91644	0.3	20000	98.0	96.4	72363	<100	101.7	99.5
Vanadium	2030	2026	0.2	2000	101.5	101.3	0	<1	99.4	96.1
Chromium	2056	2051	0.2	2000	102.8	102.6	0	<1	102.0	97.3
Manganese	2212	2200	0.5	2000	101.3	100.7	186	<5	100.8	96.6
Iron	22787	22900	0.5	20000	102.0	102.6	2386	<20	103.3	99.6
Cobalt	2057	2057	0.0	2000	102.9	102.9	0	<10	102.9	98.6
Nickel	2045	2030	0.7	2000	102.3	101.5	0	<1	102.3	98.0
Copper	2032	2020	0.6	2000	101.5	100.9	2	<1	101.5	99.1
Zinc	2025	2021	0.2	2000	100.2	100.0	22	<10	98.9	98.8
Arsenic	2034	2039	0.2	2000	101.6	101.9	2	<1	101.1	98.1
Selenium	1991	2027	1.8	2000	99.6	101.4	0	<1	102.3	97.8
Molybdenum	2029	2045	0.8	2000	101.3	102.1	3	<10	103.7	99.7
Silver	20.17	20.42	1.2	20	100.9	102.1	0	<0.2	104.4	98.1
Cadmium	1988	2008	1.0	2000	99.4	100.4	0	<0.2	103.4	98.4
Antimony	1990	2061	3.5	2000	99.4	103.0	2	<1	99.9	105.3
Barium	2065	2076	0.5	2000	101.0	101.6	45	<5	103.5	99.8
Thallium	2022	2010	0.6	2000	101.1	100.5	0	<0.5	101.3	100.2
Lead	2005	2019	0.7	2000	100.3	101.0	0	<1	101.7	99.1

* Matrix spike precision range +/- 20% RPD

** Matrix spike accuracy range +/- 30% recovery

*** LCS accuracy range +/- 15% recovery / Ind std accuracy range +/- 10% recovery

Comments: _____